

Interactive 3D Experience Spatial Theory & Level Design Diploma in Immersive Media & Game Design Year 2 (2023/24) Semester 2.1	Version 1
	Date Apr 2024
Assignment Project 2 (I3E - 30%, STLD - 20%) Designing & Development of an Interactive 3D Level	Week 6 – 11

OVERVIEW

REQUIREMENTS

- Design & document a 3D virtual level based on a scenario
- Creation of a 3D virtual level with careful consideration to User Experience
- This is an **individual** project
- All static 3D mesh assets must be built from scratch; no external assets are to be used.
- Assents are allow to use external resources are :
 1. tree/ vegetation,
 2. characters /monsters,
 3. Texture, material

INTENDED LEARNING OUTCOMES

1. Understand the concept and workflow of Level Design
2. Design & document a game level
3. Build game level with game engine and other resources
4. Utilize C# programming language and Unity engine to create a robust 3D virtual level
5. Develop and design a player experience using Unity
6. Maintaining code cleanliness using comments

1. INTRODUCTION

The Unity Engine allows developers and designers to create tangible, meaningful interactive experiences from their imaginations. It offers a wide array of tools that enable its users to create both 2D and 3D applications.

For this assignment, you will assume the role of 3D Unity Developer and use your skills to bring your idea to life. You are tasked to create a 3D virtual world that contains objects that users can interact with and achieve a meaningful result. The user must control an avatar through a first-person camera view.

The target platform for this project is Windows desktop. You are required to test all assets, scenes, and scripts are working to your needs.

SCENARIO

You have lost control of your spaceship & crashed landed on an unknown planet. You woke up, in a daze, looking around at your spaceship's bridge of your damaged spaceship. You found your way out of your spaceship to the terrain of the unknown planet. You have to explore the unknown planet to look for ways to repair your spaceship and escape from the unknown planet.

2. SUBMISSION CRITERIA

Project to be submitted via GitHub link.

All other files are to be submitted to **BrightSpace**.

Submit all deliverables as individual file and NOT as a zip file.

Note: Always backup your files using other mediums (eg. Google Drive, Dropbox, External HDD, ictspace server) throughout the assignment.

No.	Deliverable	Naming Convention	File Format
1.	Unity Project	I3E_STLD_Assg2_<Name>_Project	GitHub link
2.	Unity Build	I3E_STLD_Assg2_<Name>_Build.zip	.zip
3.	Read Me	I3E_STLD_Assg2_<Name>_ReadMe.pdf	.pdf
4.	Gameplay Video	I3E_STLD_Assg2_<Name>_Gameplay.mp4	.mp4
5	Level design document	STLD_Assg2_<Name>_LevelDesignDocument.pdf	.pdf

3. DELIVERABLE DETAILS

1. Unity Project (GitHub link)

- GitHub repository should be **public** so that tutors can access the project for marking.
- Include all project files (e.g scenes, scripts, materials, textures, packages, models, etc)
- All files should be organized in a proper file structure format

*Note: Files should be properly named and structured. Failure to do so will result in **heavy penalization** of marks. Marks are given for proper file organization. Any .rar files or other compression formats will be rejected.*

2. Unity Build

- Final build of the Unity Application

3. ReadMe

- Readme should include detailed instructions (key controls, game hack, answer key, etc) on how to use and run your application
- Indicate the **platforms/hardware/desired settings** required to run your app
- Indicate the limitations or bugs in the application
- Indicate **references and credits to the models/textures/materials** used
- Include the answers on how to solve your game puzzles if any

4. Gameplay video

- Short video of gameplay, lasting at least 1 minute.
- No elaboration of how development was done is needed.

5. Level design document

- The document should detailed everything you envisioned for the level.
- It should include Level Abstract / Constrains, Bubble Diagram, Rough Map & Asset List, Research, Design Rationale, etc...
- Should be treated as a trailer of sorts.

4. IMPORTANT TASKS & DELIVERABLES

- Read the requirements for the application carefully. Some requirements are **necessary for a Pass grade**.
- There should be a starting GUI, how to play guide on the first scene.
- For submission, you should
 - Upload the files to the assignment submission link in **BrightSpace**
 - Cross check all your files are working and can be opened. It is not uncommon that students submit shortcuts instead of the working files.
- A **readme.pdf** that describes the functions & features in your application plus a guide on how to use your application (it is highly recommended you can include a **short video guide**). Failure to provide an **understandable** readme will result in penalisation.
- All documents submitted must be in **PDF format**

DUE DATE

Assignment is due on **Week 11**.

1. ASSIGNMENT WEIGHTAGE

I3E:

ITEMS	WEIGHTAGE
Scripting & Project Management <i>Appropriate coding practises? Usage of comments? Files are organised properly? Usage of Prefabs when appropriate</i>	25%
Interactivity <i>Interactivity between player and the environment? Are the player controls implemented well? Can the interactions affect one another?</i>	25%
UI/UX <i>Are the designs of the different menus cohesive with each other and the rest of the game? Are there custom assets? Are UI elements used and placed purposefully?</i>	15%
Game Juice <i>Are the audio effects implemented well in the game? Do they change with the state of the game? What feedback mechanisms are in the game? How does the game react to player action?</i>	15%
Git Usage <i>Has Git been used for version control on the project?</i> <i>Are commits and pushes performed routinely?</i>	10%
Video & Documentation	10%
GRAND TOTAL	100%

STLD:

ITEMS	WEIGHTAGE
Level Design Document <i>Appropriate & clear documentation of the game level.</i> <i>Taking Spatial Theory into design considerations.</i> <i>All required sections are present.</i> - Level Abstract / Constrains - Bubble Diagram - Rough Map - Asset List	30%
Implementation of Level Design <i>Implementation of level using game engine.</i> <i>Use of engine tools to build the level.</i> <i>Integrating assets & other resources into the level.</i>	50%
Project Management & Progress <i>Able to submit all deliverables on time.</i> <i>Meet all checkpoints criteria.</i>	10%
Video <i>Video clearly shows the designed level implemented in Unity.</i>	10%
GRAND TOTAL	100%

VIRTUAL LEVEL REQUIREMENTS

Your 3D Virtual level is to **include the following**:

- A start menu.
- An options page accessible from the start menu.
- A credits page accessible from the start menu.
- A how-to-play page accessible from the start menu.
- A quit button on the start menu.
- There must be at least 2 different areas.
- The access point to the final area cannot be interacted with at the start of the experience.
 - E.g. a locked door, powered down teleporter, blocked off exit
- There must be at least 3 different types of collectibles in the world.
- One special collectible must be picked up in one area and placed in another.
 - E.g. picking up a power crystal in Area B, and placing it in an engine in Area A.
- There must be hazards that will damage or kill the player.
 - These hazards can be static.
- On player death, there must be a UI that asks if the player wants to return to the main menu, or restart the area.
- There must be at least one trigger area that causes some event to happen.
 - E.g. a wall collapsing and blocking the path, hazards appear to threaten the player, etc.
- Once all collectibles are collected and the special collectible is placed, the access point to the last area can be interacted with.
- There must be BGM and audio effects.

**Work from previous classes/other modules CAN be used for this assignment. But it is up to you to make sure it fits into the requirements.*

**3D assets built using ProBuilder will also count as custom assets.*

TECHNICAL REQUIREMENTS

Your development project is to **include the following**:

- Usage of GitHub Desktop to manage the project.
- Usage of GameManager to manage progress of the game.
- Usage of spawning mechanics.
- Usage of child classes and virtual functions.
- Usage of Unity's Physics system for interactions and triggers.
- Utilization of XML comments ("///") for ALL variables and functions.
- Utilization of class headers for ALL classes.
 - Samples of comments and headers will be found on BrightSpace.

MARKING SCHEME

To obtain this grade...	Minimally, you need to implement...
Pass	• Adequate usage of user inputs • Adequate usage of models • Models are textured • UI functions perform their expected tasks. • Interactions are done with collisions and triggers • Accessing of at least 2 areas from access points ◦ Areas can be in the same scene. • At least 3 different types of collectibles that can be collected • Final access point is locked until: ◦ Special collectible is placed • At least one trigger that causes an event. • At least one type of hazard that kills the player • At least BGM and primary sound effects implemented ◦ Primary sound effects will consist of collecting collectibles, dying, interacting with access points, any major event(s) in the game. • A congratulatory message pops up after the player clears the final area. • Script files are properly commented with file headers • Git with final project update in it
Merit Grade	You must fulfil the criteria of obtaining a passing grade, plus: • Player is able to pause the game. ◦ Pausing the game allows the player to: ▪ Restart the level ▪ Exit to main menu • All areas are in different scenes. • All interactions are done via raycasting and player input. • Use of Physic Layers and Materials • More than one use factor of triggers. • More than one type of hazard. • Hazards deal damage over time to the player instead of kill the player on one hit. • Secondary sound effects implemented • Usage of Inheritance in scripts • Git have 3 commit with good Summary and Description messages
Excellent Grade	You must fulfil the criteria of obtaining a Merit grade, plus: • There are transitions when moving from area to area. • Different types of hazards deal different amounts of damage to the player. • All actions have sound effects implemented. • Custom UI assets & models. • Usage of Polymorphism in scripts • Git have Weekly Commit (excluding Holiday)

6. SUBMISSION DEADLINE

Deadline: 2359, Sunday of Week 11 (30/06/2024)

All deliverables to be properly named and submitted on **BrightSpace** as instructed by your lecturer. Further instructions will be given closer to the deadline.

CODE REVIEW

As part of your assessment, you may be required to give a brief explanation of selected pieces of code written by you in the assignment.

PLAGIARISM AND COPYRIGHT

Plagiarism means, *“copying any part of a source, and then submitting it, claiming that it is your own work.”*

Please ensure that all the works submitted by you are not copied from other sources. Any attempt to plagiarize will be dealt with severely, and it may result in your failing the module.

- This is an **individual** assignment. Do not copy code from your classmates or online sites. All copied code must be attributed and credited
- Requirements implemented with the help of third party code will not be counted as implemented by you, and may result in loss of marks.
- If you are found committing plagiarism, you will score 0 for your assignment

LATE SUBMISSION

Late submission will be **penalised** (10% of the marks for each day late after 12 noon). Submission will **not be accepted after 5 days** (including weekends and public holidays) from the date of submission.

DUE DATE

Assignment is due on **Week 11, Sunday @1159pm.**

End of Project Brief (Total: 12 pages)